

AMLD Security 3

Eight Formal Application Sketches for
Intelligent Target Settlement

A complementary companion to
Security by Separation, Verification, and Settlement

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The older treatise develops the defensive foundations: separation, identity, threat models, containment, formal verification, the Ocean gate, and verifier-governed AMLD memory. This companion begins where that book deliberately stops. It asks how the same architecture could be applied, cautiously and formally, when an intelligent target seeker is aimed at cryptographic, software, scientific, engineering, infrastructure, or AI-safety questions. Each chapter moves from motivation to an explicit mathematical sketch, a modest proposition or conjecture, and a boundary statement. None of the sketches establishes that a deployed system is vulnerable or that AMLD has already achieved the proposed capability.

Companion Statement

This manuscript is complementary in a precise sense. The older sibling, *Security by Separation, Verification, and Settlement* [5], moves from elementary security concepts toward AMLD. The present manuscript assumes those concepts and moves outward from AMLD toward eight possible application domains. The two books therefore answer different questions:

	Older sibling	This companion
Primary question	What defensive architecture should exist before powerful reasoning is trusted?	What could verifier-gated target settlement plausibly do in specific domains?
Direction	Security foundations → AMLD	AMLD → application sketches
Main result type	Defensive theorem	Conditional application theorem or research conjecture
Recurring warning	A verifier is only as good as its model and specification.	A formal pattern is not evidence of an achieved capability.

The shared writing rule remains *motivation first, formalism immediately after*. The shared security policy remains defensive: the mathematics is organized around auditing, containment, certification, repair, and controlled deployment, not around operational exploitation.

Abstract

An automated target prover, theorem prover, or metalogical decider becomes security-relevant when its target is a vulnerability, an invariant, a compressed state representation, or a certificate about a protected system. The central problem is not merely computational power. It is how to separate creative search from the authority to promote claims into trusted state. This manuscript gives a common verifier-gated formalism and applies it to eight domains: cryptography, vulnerability research, critical infrastructure, military and intelligence compartmentation, software correctness, scientific discovery, engineering optimization, and AI safety. Across the applications, four motifs recur: certificate-level settlement by the *P/R/I/C* roles; separation of verified **BANK** state from speculative **FUTUREBANK** state; future-sufficient quotienting; and Ocean-style regression gates for candidate machinery. The results are intentionally conditional. They identify mathematical forms that a defensible application could take and expose the preservation obligations that would have to be proved. They do not report a new cryptanalytic attack, a universal vulnerability finder, or a proof of global AI safety.

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Part I

The Shared Formal Spine

Chapter 1

From Intelligent Target Seeking to Verified Settlement

1.1 Targets, certificates, and environments

Let a target instance be

$$\mathfrak{I} = (c, \Gamma, \mathcal{E}, V, \mathcal{B}),$$

where c is a claim or goal, Γ is the admitted theory or specification, $\mathcal{E}$ is the declared environment, V is an independent verifier, and $\mathcal{B}$ is a resource budget. Search may be learned, reflective, randomized, federated, or otherwise adaptive. Validity is narrower:

$$\text{Accept}(\pi, c \mid \Gamma, \mathcal{E}) \iff V(\pi, c, \Gamma, \mathcal{E}) = 1.$$

Changing a search policy may change which certificate is found and how quickly it is found; it must not silently change the acceptance relation.

1.2 Five operational outcomes

For a formal target c , an AMLD-style process distinguishes

$$\text{Settle}(c) \in \{P, R, I, C, \text{UNKNOWN}\}.$$

Here P supplies a verified certificate for c , R supplies one for $\neg c$, I supplies an independence certificate in a declared metatheory, and C supplies an inconsistency certificate for the admitted assumptions. The fifth outcome records resource-bounded nonsettlement. It is crucial that

$$\text{UNKNOWN} \not\vdash c, \quad \text{UNKNOWN} \not\vdash \neg c, \quad \text{UNKNOWN} \not\vdash I(c).$$

1.3 Epistemic separation

Let B_t be verified shared state and F_t speculative state at time t . Candidate objects may be generated freely into F_t , but promotion is restricted:

$$B_{t+1} = B_t \cup V_t(A_t), \quad V_t(A_t) := \{a \in A_t : V(a, \Gamma_t, \mathcal{E}_t) = 1\}.$$

This is the mathematical core of BANK/FUTUREBANK separation. Provenance is retained, but provenance does not substitute for verification.

Proposition 1.1 (Verifier-gated noncontamination). *Suppose the displayed update is the only rule that adds semantic assertions to B_t . Then every new assertion in $B_{t+1} \setminus B_t$ has been accepted by V_t .*

Proof. By the update rule, $B_{t+1} \setminus B_t \subseteq V_t(A_t)$. Membership in $V_t(A_t)$ is defined by verifier acceptance. $\square$

1.4 Future-sufficient quotienting

Let $\mathcal{H}$ be a set of search histories and $Q : \mathcal{H} \rightarrow \mathcal{S}$ a descriptor. The equivalence relation $h_1 \sim_Q h_2$ iff $Q(h_1) = Q(h_2)$ is computationally useful only if it preserves the future facts relevant to the target. One strong form is

$$Q(h_1) = Q(h_2) \implies \text{Future}(h_1) \equiv_V \text{Future}(h_2),$$

where $\equiv_V$ means equality of verifier-relevant continuations, dependency obligations, scope, and certificate-lifting behavior.

Proposition 1.2 (Sound quotient reuse). *Assume each quotient certificate $\bar{\pi}$ accepted at state $Q(h)$ has a refinement map L_h with*

$$\bar{V}(\bar{\pi}, Q(h)) = 1 \implies V(L_h \bar{\pi}, h) = 1.$$

Then reusing accepted quotient computation cannot introduce a false promoted claim in the original state space.

Proof. Every promotion from a quotient state is refined by L_h and checked by V . Soundness is therefore inherited from the stated lifting condition, not from the quotient heuristic itself. $\square$

1.5 Guidance and the Ocean gate

A COMPASS-type score may prioritize actions,

$$S(a \mid h) = w^\top \phi(a, h) - \lambda \widehat{\text{Cost}}(a \mid h) - \mu \widehat{\text{Risk}}(a \mid h),$$

but S is not a truth predicate. Candidate search machinery M can itself be tested against a versioned Ocean suite $\mathcal{O}_t$. If $I_1, \dots, I_k$ are hard invariants, a conservative gate is

$$\text{Acc}(M, \mathcal{O}_t) = 1 \iff \bigwedge_{o \in \mathcal{O}_t} \bigwedge_{j=1}^k I_j(M, o) = 1.$$

A verified failure r_t may then be retained by $\mathcal{O}_{t+1} = \mathcal{O}_t \cup \{r_t\}$. Passing a finite gate excludes tested failures; it does not prove universal safety.

Chapter 2

Application Map

Domain	Formal target	Useful certificate	Central preservation obligation
Cryptography	Hardness-relevant invariant or quotient	Proof that a reduction or lifting map preserves the target problem	Do not merge histories with different verifier-relevant futures
Vulnerability research	Conformance of implementation to policy	Checked violating trace or inductive proof	Model the complete sufficient routes and system state
Critical infrastructure	Bounded compromise propagation	Reachability certificate and non-regression record	Containment must not destroy required availability
Secure domains	Authorized boundary crossing and noninterference	Invariant proof or violating trace	Preserve labels, authority, provenance, and declassification rules
Software correctness	Safety invariant over reachable states	Inductive proof or finite counterexample	Checker and dependency closure remain fixed
Scientific discovery	Predictively sufficient low-dimensional state	Theorem, checked derivation, or preregistered empirical certificate	Preserve the declared observable family, not merely training fit
Engineering optimization	Feasible and objective-preserving state merge	Correspondence between feasible continuations	Preserve downstream constraints and optimum values
AI safety	Separation of generation, verification, testing, and authority	Gate receipts and invariant certificates	Policy revision must not revise its own truth conditions silently

The table is not evidence that all eight applications are equally mature. It identifies the object that would need to be certified before speculative target seeking could be treated as defensive knowledge.

Part II

Eight Plausible Applications

Chapter 3

Cryptography

Discovering unexpected structure in a problem believed to be hard could matter to cryptography. The defensible research question is not whether AMLD can simply try keys faster. It is whether a small, independently checkable invariant or quotient can replace a much larger search without changing the cryptographic problem being solved.

3.1 Plausible application

Let X_κ be an instance family at security parameter κ , let $\mathcal{H}_\kappa$ be legal search histories, and let $T(h, a)$ be continuation by action a . A quotient candidate $Q_\kappa : \mathcal{H}_\kappa \rightarrow \mathcal{S}_\kappa$ induces

$$h_1 \sim_Q h_2 \iff Q_\kappa(h_1) = Q_\kappa(h_2).$$

For defensive significance, equality of quotient states must imply more than similar local features. It should preserve terminal status, admissible continuations, certificate dependencies, and success probability under the declared model. A strong deterministic obligation is a verifier-respecting bisimulation:

$$h_1 \sim_Q h_2 \implies \begin{cases} V_{\text{term}}(h_1) = V_{\text{term}}(h_2), \\ \forall a_1 \in A(h_1) \exists a_2 \in A(h_2) : T(h_1, a_1) \sim_Q T(h_2, a_2), \\ \forall a_2 \in A(h_2) \exists a_1 \in A(h_1) : T(h_1, a_1) \sim_Q T(h_2, a_2). \end{cases}$$

If the search is stochastic, the corresponding condition is equality of the pushed-forward transition kernels on $\mathcal{S}_\kappa$. A COMPASS-like policy may search over candidate descriptors by a score

$$J(Q) = \widehat{\text{reuse}}(Q) - \lambda \log |\text{im } Q| - \mu \widehat{\text{Cost}}(Q) - \nu \widehat{\text{collision risk}}(Q),$$

but promotion requires a certificate for the preservation obligation. Under that certificate, dynamic programming or lemma reuse may occur once per quotient class rather than once per history. The scientific object is therefore a proved structural compression of an admitted problem, not an unverified claim that a named encryption scheme is weak.

Conjecture 3.1 (Certified quotient advantage). *For some structured cryptographic audit families, there exists a future-sufficient Q for which verified quotient search uses asymptotically fewer distinct*

continuation states than history-level search while returning only liftable certificates.

Boundary note 3.2. No argument in this chapter establishes an efficient attack on AES, RSA, elliptic-curve cryptography, or a post-quantum construction. Hardness is scheme-, model-, and parameter-specific.

Source anchors. Safe quotienting and COMPASS are developed in *What Checks the Proof?* v6.80, pp. 28–29 and 68 [6]; security as a settlement problem appears in the older security treatise, pp. 42–43 [5].

Chapter 4

Cybersecurity and Vulnerability Research

A target-seeking system could generate many suspected traces. Defensively, the important move is to replace the vague target “find a vulnerability” by a model-relative conformance claim and a machine-checkable defect certificate. Speculation remains useful, but it remains quarantined.

4.1 Plausible application

Let $\mathcal{M} = (S, S_0, A, \rightarrow)$ be a labeled transition system and let Z be its complete finite or infinite traces. A policy defines $\text{Req}(z) \in \{0, 1\}$; the implementation semantics defines $\text{Impl}(z) \in \{0, 1\}$. One-sided acceptance soundness is

$$\Phi_{\text{safe}} := \forall z \in Z [\text{Impl}(z) \Rightarrow \text{Req}(z)].$$

If a finite trace $z^* = (s_0, a_0, \dots, s_n)$ carries receipts proving

$$s_0 \in S_0, \quad s_i \xrightarrow{a_i} s_{i+1}, \quad \text{Impl}(z^*) = 1, \quad \text{Req}(z^*) = 0,$$

then z^* is a finite R -certificate. Conversely, a P -certificate may consist of an inductive invariant I satisfying

$$S_0 \subseteq I, \quad (s \in I \wedge s \rightarrow s') \Rightarrow s' \in I, \quad s \in I \Rightarrow \neg \text{Bad}(s).$$

The I -role asks a different question: whether Φ_{safe} is independent of the admitted audit theory in a declared metatheory. The C -role searches for inconsistent policy obligations. Candidate traces, invariants, and repairs reside in F_t ; only checker-accepted objects enter B_t . For an evolving analyzer M_t , Ocean admission can require preservation of four architectural invariants: rejected candidates never enter B_t ; every derived record has a receipt; dependencies are closed; and resource limits are enforced. This converts vulnerability research into an evidence-producing audit process in which the reasoning system proposes and prioritizes, while a separate checker determines what has actually been established.

Proposition 4.1 (Counterexample sufficiency). *If the displayed receipts for z^* verify, then Φ_{safe} is*

false in the declared transition model and policy.

Proof. The verified trace instantiates the universal quantifier with an implemented acceptance that is not required, falsifying the implication. □

Boundary note 4.2. A checked defect is relative to a model and specification. An incomplete model can miss real behavior; an inadequate policy can be implemented perfectly and still protect the wrong property.

Source anchors. The authentication-defect certificate and four-role audit appear in *What Checks the Proof?* v6.80, pp. 54–55 [6]; the Ocean moat appears in the older treatise, pp. 32–33 [5].

Chapter 5

Critical Infrastructure

For hospitals, power systems, water utilities, transportation, and communications, an initial cyber failure can become a physical or availability failure. The natural defensive object is therefore not one secure/insecure bit but a graph of propagation, authority, service, and recovery.

5.1 Plausible application

Model the system as a directed labeled graph $G = (V, E, \ell, w)$, where $\ell(e)$ records the type of transition and $w(v) \geq 0$ is a declared consequence weight. For a compromised component c , define

$$B_G(c) = \text{Reach}_G(c), \quad L_G(c) = \sum_{v \in B_G(c)} w(v).$$

A segmentation design D produces $G_D = (V, E_D)$ together with a set R_D of service paths that must remain available. If $E_{D_2} \subseteq E_{D_1}$, reachability monotonicity gives

$$B_{G_{D_2}}(c) \subseteq B_{G_{D_1}}(c), \quad L_{G_{D_2}}(c) \leq L_{G_{D_1}}(c).$$

But removal of every edge is not a useful defense. Let

$$\mathbf{J}(D) = \left(p_{\text{comp}}(D), \max_c L_{G_D}(c), A_{\text{loss}}(D), C_{\text{verify}}(D), C_{\text{repair}}(D) \right).$$

A candidate dominates another only under a declared partial order and hard service constraints, for example

$$D_2 \succ_X D_1 \iff m_X(D_2) < m_X(D_1) \wedge \forall j \in J_{\text{protected}}, m_j(D_2) \leq m_j(D_1) \wedge R_{D_2} \models \Psi_{\text{service}}.$$

An AMLD controller can let P prove containment and service obligations, R find checked violating paths, I test whether the obligations are independent of the model, and C detect incompatible requirements. The useful output is a certificate-bearing Pareto comparison, not a single heuristic security score.

Proposition 5.1 (Edge-removal containment). *If $E' \subseteq E$, then for every $c \in V$, $\text{Reach}_{(V, E')}(c) \subseteq \text{Reach}_{(V, E)}(c)$.*

Proof. Every path using only edges in E' is also a path using edges in E . □

Boundary note 5.2. The proposition proves containment in the graph model only. It does not validate the graph, estimate compromise probability, or show that required services survive segmentation.

Source anchors. Blast radius and narrow bridges are treated in the older security treatise, pp. 10–11; multi-objective security and separation-aware control appear on pp. 53–56 [5].

Chapter 6

Military and Intelligence Compartmentation

Compartmentation gives a mathematically clean example of the book's security philosophy: separate domains, restrict crossings, preserve provenance, and never grant speculative reasoning direct operational authority. The formulation below is about defensive authorization and noninterference, not operational attack planning.

6.1 Plausible application

Let $(L, \preceq)$ be a security-label lattice and let each state component x carry label $\lambda(x) \in L$. Partition the state space into domains

$$V = \bigsqcup_{i=1}^k V_i.$$

For every cross-domain transition $e = (x, y)$, define a boundary predicate $P_e(s, \alpha, \rho)$, where s is current state, α is asserted authority, and ρ is provenance. The safety target is

$$\Phi_{\text{cross}} := \Box (\text{Fire}(e, s, \alpha, \rho) \Rightarrow P_e(s, \alpha, \rho)).$$

For confidentiality, let $s \equiv_\ell s'$ mean that states agree on observations visible at low label ℓ . A possibilistic noninterference target is

$$s_0 \equiv_\ell s'_0 \Longrightarrow \text{Obs}_\ell(\text{Run}(s_0, u)) = \text{Obs}_\ell(\text{Run}(s'_0, u))$$

for every admitted low action sequence u , except at explicitly modeled declassification gates. Role P may prove these invariants compositionally; R may supply a checked boundary-crossing or observational counterexample; I and C retain their metatheoretic meanings. Search output does not become trusted merely because a designated role produced it:

$$c \in B_{t+1} \setminus B_t \Longrightarrow \exists \pi V(\pi, c, \Gamma, \mathcal{E}) = 1.$$

Thus compartmentation is represented simultaneously as a graph boundary, an authorization predicate, and an observational equivalence, with the verifier mediating any promotion into trusted state.

Conjecture 6.1 (Cross-domain shared-certificate advantage). *For some compartmented systems, a provenance-preserving shared BANK of verified interface lemmas reduces total audit cost relative to isolated audits without weakening domain-local acceptance.*

Boundary note 6.2. Noninterference depends on the observation model, declassification rules, scheduler, and environment. The displayed formula is not a universal characterization of national-security systems.

Source anchors. Separation, least privilege, and narrow bridges are treated in the older security treatise, pp. 9–11; *P/R/I/C* security settlement appears on pp. 42–43 [5].

Chapter 7

Software Correctness

Software correctness is the most direct beneficial application because both proof and refutation can often be represented by finite machine-checkable objects. Learned guidance may choose what to try; it need not be trusted to decide what counts as a valid proof.

7.1 Plausible application

Let $\mathcal{M} = (S, S_0, \rightarrow)$ be a transition system and let $I : S \rightarrow \{0, 1\}$ be a safety invariant. An inductive certificate proves

$$\forall s \in S_0 \ I(s), \quad \forall s, s' \in S \ [I(s) \wedge s \rightarrow s'] \Rightarrow I(s').$$

It follows that I holds throughout $\text{Reach}_{\mathcal{M}}(S_0)$. Role P searches for such an invariant, auxiliary lemmas, and checker receipts. Role R searches for a finite trace

$$s_0 \rightarrow s_1 \rightarrow \cdots \rightarrow s_n, \quad s_0 \in S_0, \quad \neg I(s_n).$$

The reasoning architecture is also software and should satisfy its own executable invariants:

$$u \in B_{t+1} \setminus B_t \Rightarrow \exists r \ V(r, u) = 1,$$

$$V(r, u) = 0 \Rightarrow u \notin B_{t+1}, \quad u \in B_t \Rightarrow \text{Dep}(u) \subseteq B_t.$$

If verifier receipts are content-addressed and dependencies are immutable, replay can check that the accepted object still means what it meant at promotion time. Candidate proofs and candidate program repairs remain in F_t until checking succeeds. An Ocean suite then includes historical counterexamples, malformed certificates, resource-limit tests, and dependency-closure cases. This creates two mutually reinforcing proof obligations: the application program must preserve its safety invariant, and the settlement machinery must preserve the epistemic boundary by which that claim is accepted.

Proposition 7.1 (Inductive reachability safety). *If the base and step obligations above hold, then every reachable state satisfies I .*

Proof. Induct on the length of a path from S_0 . The base case uses $S_0 \subseteq I$; the step uses transition

preservation. □

Boundary note 7.2. The theorem establishes only the invariant encoded by I in the modeled semantics. It does not prove that the model matches compiled code, hardware, operators, or every environmental condition.

Source anchors. Policy-neutral verification and executable architecture invariants appear in *What Checks the Proof?* v6.80, pp. 10 and 69–70 [6].

Chapter 8

Scientific Discovery

The Quotient Hunter idea is scientifically interesting when many apparent histories can be replaced by a smaller state that preserves everything relevant to future prediction or verification. Invariant discovery then becomes a search for sufficient representation, not merely a search for correlation.

8.1 Plausible application

Let H_t denote an observed history and $Y_{t:\infty}$ a declared family of future observables. A descriptor $Q(H_t)$ is predictively sufficient when

$$\mathcal{L}(Y_{t:\infty} \mid H_t) = \mathcal{L}(Y_{t:\infty} \mid Q(H_t)).$$

Equivalently, under regularity conditions,

$$Y_{t:\infty} \perp H_t \mid Q(H_t).$$

For deterministic models and observable family $\mathcal{F}$, the obligation may instead be

$$Q(h_1) = Q(h_2) \Rightarrow \forall f \in \mathcal{F} \forall \tau \geq 0, \quad f(Y_{h_1}(\tau)) = f(Y_{h_2}(\tau)).$$

Candidate descriptors can be ranked by an information-compression objective

$$U(Q) = \widehat{I}(Q(H_t); Y_{t:\infty}) - \lambda I(Q(H_t); H_t) - \mu \widehat{\text{Cost}}(Q),$$

or by an exact symbolic compression metric when a formal model is available. Yet $U(Q)$ is only a search score. A descriptor enters verified state only after a domain-appropriate certificate: a proof in a formal model, an exact algebraic derivation, or a preregistered empirical test with a clearly limited conclusion. FUTUREBANK may retain failed invariants and conditional “what if” models without asserting them. If a descriptor passes the preservation gate, downstream inference may reuse the same reasoning for every history in a quotient class. The payoff is a certified reduction of effective state dimension and an explicit statement of which observables survive the reduction.

Conjecture 8.1 (Verifier-history discovery advantage). *On some repeated scientific-model families, retaining rejected quotient hypotheses with provenance reduces expected rediscovery cost without*

increasing false certification, relative to retaining only accepted descriptors.

Boundary note 8.2. Predictive sufficiency is relative to the chosen observables and data-generating assumptions. Good holdout prediction alone is not a theorem that the quotient is universally correct.

Source anchors. FUTUREBANK, planning, and COMPASS appear in *What Checks the Proof?* v6.80, pp. 27–30; future-sufficient quotienting appears on p. 68 [6].

Chapter 9

Engineering Optimization

Scheduling, placement, routing, circuit design, materials design, and configuration all contain large combinatorial spaces. Safe state merging can reduce repeated work only when it preserves future constraints and objective values.

9.1 Plausible application

Consider

$$\min_{x \in X} f(x) \quad \text{subject to} \quad g_i(x) \leq 0 \ (i = 1, \dots, m), \quad h_j(x) = 0 \ (j = 1, \dots, r).$$

For a partial design history h , let

$$\mathcal{C}(h) = \{x \in X : x \text{ extends } h \text{ and satisfies inherited constraints}\}.$$

A strong exact equivalence is

$$h_1 \sim h_2 \iff \exists \varphi : \mathcal{C}(h_1) \rightarrow \mathcal{C}(h_2) \text{ bijective such that } f(x) = f(\varphi(x))$$

and all checker-relevant labels are preserved. A weaker quotient may preserve only feasibility and an optimal value:

$$Q(h_1) = Q(h_2) \Rightarrow [\mathcal{C}(h_1) = \emptyset \iff \mathcal{C}(h_2) = \emptyset] \wedge \inf_{x \in \mathcal{C}(h_1)} f(x) = \inf_{x \in \mathcal{C}(h_2)} f(x).$$

Under either certified obligation, dynamic programming may cache downstream value and feasibility by quotient state. A COMPASS-like scheduler may prioritize

$$S(a \mid h) = \alpha \Delta_{\text{goal}} + \beta \rho_Q + \gamma \text{reuse} - \delta \widehat{\text{Cost}} - \eta \widehat{\text{Risk}},$$

while feasibility remains checker-defined:

$$\text{Feas}(x) \iff \bigwedge_i g_i(x) \leq 0 \wedge \bigwedge_j h_j(x) = 0 \wedge V(x, \Gamma) = 1.$$

Learned guidance therefore decides where to spend computation; it does not redefine admissibility. The research question is whether certified equivalence classes can be discovered cheaply enough that their reuse benefit exceeds the cost of proving them.

Proposition 9.1 (Optimal-value reuse). *If Q satisfies the displayed optimal-value preservation obligation, then an exact cached value at $Q(h)$ is valid for every history in that quotient class.*

Proof. The obligation states equality of the infimum for any two representatives of the same class. $\square$

Boundary note 9.2. Preserving an optimum value does not necessarily preserve an optimizer, robustness margin, or implementation cost. Those quantities must be added to the equivalence obligation when relevant.

Source anchors. COMPASS and safe quotienting appear in *What Checks the Proof?* v6.80, pp. 28–29 and 68 [6].

Chapter 10

AI Safety Itself

A shortcut-generating system is materially different from a shortcut-generating system that is also allowed to declare its own outputs valid. The application here is architectural: separate generation, testing, verification, memory promotion, and operational authority.

10.1 Plausible application

Let G_θ be a mutable generator or search policy and V_ω a separately governed verifier. For a claim c , certificate π , theory Γ , and environment $\mathcal{E}$, define

$$\text{Promote}(c, \pi) \iff V_\omega(c, \pi, \Gamma, \mathcal{E}) = 1.$$

Policy-neutral verification requires generator changes not to alter acceptance:

$$\forall \theta_1, \theta_2, c, \pi, \quad V_\omega(c, \pi, \Gamma, \mathcal{E}) \text{ is independent of whether } G_{\theta_1} \text{ or } G_{\theta_2} \text{ proposed } (c, \pi).$$

Candidate self-modifications θ' remain speculative,

$$\theta' \in F_t \not\Rightarrow \theta' \in B_t,$$

and must pass hard invariants $H_1, \dots, H_k$ plus an Ocean regression suite:

$$\text{Gate}(\theta') = 1 \iff \left(\bigwedge_{o \in \mathcal{O}_t} \bigwedge_{j=1}^k H_j(G_{\theta'}, o) = 1 \right) \wedge V_\omega(\pi_{\theta'}, \theta', \Gamma_{\text{deploy}}) = 1.$$

The gate may also include a statistical endpoint, but statistical success cannot override a failed hard invariant. A verified regression failure r_t updates $\mathcal{O}_{t+1} = \mathcal{O}_t \cup \{r_t\}$. Operational authority is granted by a distinct deployment controller with least privilege and rollback, not by membership in a speculative search frontier. This does not solve alignment in general. It supplies a checkable separation law: policy may change attention, but promotion remains conditional on an independently governed acceptance relation.

Proposition 10.1 (Generator-independent promotion). *Under the displayed promotion rule, chang-*

ing θ alone cannot cause a certificate rejected by V_ω to enter B_t through the modeled update.

Proof. Promotion depends on V_ω , not on the identity or parameters of the proposing generator. Apply Proposition 1.1. $\square$

Boundary note 10.2. Independent verification is a safety mechanism, not a proof of global AI safety. The verifier, specification, interfaces, deployment controller, and environment remain possible failure surfaces.

Source anchors. Policy-neutral verification and safe quotienting appear in *What Checks the Proof?* v6.80, pp. 10 and 68 [6]; the Ocean moat appears in the older security treatise, pp. 32–33 [5].

Part III

Research Boundaries and Program

Chapter 11

A Cross-Domain Experimental Program

The eight sketches can be tested without pretending that they are already established capabilities. For each application family, freeze

(target family, model, verifier, budget, baseline, split, stopping rule, metric).

Compare at least four conditions: unguided certified search; guided search without quotient reuse; guided search with candidate quotienting but no promotion; and guided search with verifier-certified quotient reuse. A useful primary endpoint is verified settlement per unit resource. Secondary endpoints include certificate size, wall time, peak memory, repeated-search cost, false-promotion count, and non-regression coordinates such as availability.

11.1 A minimum preregistration schema

1. State the target claim and the exact verifier acceptance relation.
2. State which data or instances inform search and which remain held out.
3. State the quotient preservation obligation before observing target results.
4. Match baselines on verifier, budget, and stopping conditions.
5. Record every promoted object with its certificate, dependencies, provenance, and environment.
6. Treat UNKNOWN as a distinct outcome rather than silently reclassifying it.
7. Retain verified failures as regression cases and report all non-regression coordinates.

Conjecture 11.1 (Separation-aware advantage). *On some repeated target-settlement distributions, a controller that optimizes both settlement progress and verifier-boundary preservation attains a better Pareto frontier than a matched controller that optimizes settlement success alone.*

Chapter 12

Claims This Manuscript Does Not Make

1. No finite Ocean suite proves universal security.
2. No heuristic score is promoted to a truth predicate.
3. No failure to settle is treated as proof of safety, falsity, or independence.
4. No quotient is safe merely because it compresses a benchmark well.
5. No proof of implementation conformance establishes adequacy of the specification.
6. No component-local proof automatically establishes secure composition.
7. No chapter reports a new practical cryptanalytic attack or operational vulnerability.
8. No verifier-gated architecture, by itself, proves global AI safety.

The positive claim is narrower and, for research, more useful: each application can be formulated so that the powerful component proposes search directions while an explicit preservation obligation and independent checker govern promotion. That is the precise point at which this companion joins its older sibling.

Appendix A

Glossary

AMLD

A verifier-gated, multi-role architecture for settling formal targets through proof, refutation, independence, contradiction, or an explicit resource-bounded unknown outcome.

Blast radius

The set or weighted consequence of resources reachable after a modeled compromise.

Certificate

A finite object whose validity for a declared claim can be checked independently.

COMPASS

Learned or engineered guidance that ranks search actions without defining truth.

Future-sufficient quotient

A state compression that identifies histories only when all declared future facts needed for search and verification are preserved.

Ocean

A versioned adversarial and regression environment used before candidate machinery is granted trusted authority.

Policy-neutral verification

The requirement that changing search policy does not silently change the acceptance relation.

Settlement

Acceptance of a certificate for one of the declared $P/R/I/C$ outcomes.

Verified BANK

Shared trusted state whose semantic assertions enter only through the declared verification gate.

FUTUREBANK

Quarantined speculative state containing hypotheses, imagined traces, candidate quotients, and proposed repairs not yet admitted to the verified BANK.

Appendix B

Symbols and Notation

Symbol	Meaning in this manuscript
c	Target claim or goal
Γ	Admitted theory, specification, and scoped dependencies
$\mathcal{E}$	Declared environment or threat model
V	Independent verifier or checker
P, R, I, C	Prover, Refuter, Independence, and Contradiction settlement roles
UNKNOWN	No accepted settlement certificate within declared conditions
B_t	Verified BANK contents at time t
F_t	FUTUREBANK speculative contents at time t
Q	Candidate future-sufficient descriptor or quotient map
$h_1 \sim_Q h_2$	Histories have the same quotient state
$\mathcal{O}_t$	Ocean regression suite at time t
$\text{Reach}_G(c)$	Vertices reachable from c in graph G
$D_2 \succ_X D_1$	D_2 strictly improves the declared metric for vulnerability class X , subject to stated non-regression constraints

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